

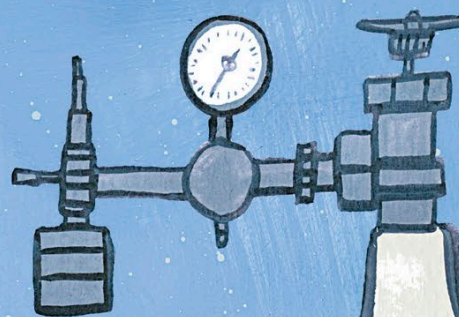


Separating oxygen from air allowed it to be put in tanks and transported anywhere in the world.

Refrigeration innovation

Carl was especially curious about how machines might be used to cool things down. In the late 1800s, most factories, homes, and stores still relied on blocks of ice cut from frozen rivers and lakes to keep things cold. Carl set out to improve existing cooling systems and ended up inventing some of the world's most effective refrigeration machines.

He gave up his teaching job and set up a company to build them for all types of customer—from food makers and dairies, to factories and science labs. However, Carl didn't stop there. He then discovered how to use cooling systems to separate out the different gases in air—mainly oxygen—in quantities big enough to sell. Each gas has different uses when separated from the others. This breakthrough changed the world, because oxygen could now be stored in tanks and used when needed, such as for patients in hospitals.



The chemical formula for oxygen is O_2 .

Pure oxygen has many uses. It can be found in hospitals around the world, and it is even used to burn fuel in space rockets!

OXYGEN

Liquefying air

By squeezing air then letting it expand over and over again, Carl was able to cool it down enough so it became liquid. Each of the gases in air changes back from a liquid to a gas at a different temperature. By warming the liquid air very slowly, Carl was able to separate the gases, including oxygen, out one by one and put them into tanks. He made machines that could do this with huge amounts of air.

At first, pure oxygen was used in powerful oxyacetylene torches. These burn with a flame so hot that they can melt and cut through metal.

The flame comes out of this end.

These torches made it easier to build skyscrapers, ships, and other steel structures.